Geography-Aware Province-Level Vietnamese Dialect Identification with Frozen Pretrained Speech Encoders

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***Abstract*—We study province-level Vietnamese dialect identification on ViMD, a 102-hour corpus of interviews broadcast by the television stations of all 63 provinces. Instead of fine-tuning, we keep two pretrained speech encoders frozen, the self-supervised XLS-R and the ASR-trained PhoWhisper, and train a light head over a learned weighted sum of their layers. A head over fused XLS-R and PhoWhisper representations reaches 63.9% accuracy as a three-seed ensemble (61.0% for individual seeds), and an ensemble of nine heads reaches 64.6% accuracy and 0.637 macro-F1 on the official test set; the best fine-tuned baseline reported with the dataset obtained 0.411 macro-F1 under a different protocol. We also study Geo-aware Label Smoothing (Geo-LS), an instance of similarity-based label smoothing with a geographic kernel: it keeps a fixed mass on the true province and spreads the smoothing mass to geographically close provinces. On the fused model, Geo-LS lowers the mean distance between the true and predicted province by 8.2 km (95% speaker-level CI 1.2 to 15.3) relative to standard label smoothing; the accuracy difference is -0.3 points, and its 90% interval (-1.3 to 0.7) lies within ± 1.5 points. Control analyses show that the accuracy is largely retained without spoken place names and on utterances that text baselines misclassify, and that it does not differ significantly between male and female speakers. Robustness tests, however, show that the system depends on the recording conditions: equalising each utterance's long-term spectrum lowers accuracy from 64% to 47% (restoring the true province's average spectrum instead recovers about 9 points), and 20 dB white noise or telephone-band audio lowers it to about 30% and 16%. Because each label denotes the province of the broadcasting station, the results describe broadcast recordings and do not establish speaker-origin identification from arbitrary audio. In the configurations we tried, full fine-tuning, LP-FT and temporal CNN or Transformer heads did not outperform the frozen-feature approach in this low-resource setting of about 240 training utterances per province.**

***Index Terms*—dialect identification, Vietnamese, self-supervised learning, XLS-R, Whisper, label smoothing, geographic distance**

I. Introduction

Vietnamese is usually described with three dialect regions, Northern, Central and Southern, but pronunciation also varies from province to province. Knowing a speaker's dialect helps speech recognition, speaker profiling and sociolinguistic research. The ViMD corpus [1] made the fine-grained task measurable by labelling speech from all 63 provinces. Its labels give the province of the broadcasting station, used as a proxy for the speaker's provincial dialect (Section III). Following the dataset paper we call the task dialect identification, but strictly our models classify the broadcasting province of a recording, and we evaluate how much of that classification reflects accent rather than recording conditions. Its baselines fine-tune wav2vec 2.0 and Whisper models and reach at most 0.411 macro-F1.

The task has two properties that standard classification ignores. First, the classes have a geography: confusing Hà Nội with Hà Nam is a much smaller error than confusing it with Cà Mau. Second, a province label can be predicted from cues that are not dialect at all, such as place names spoken in the recording or the acoustic signature of the local broadcaster that recorded it. A useful study has to measure both.

This paper asks four research questions:

RQ1 How far can frozen pretrained speech representations go on 63-way dialect identification?

RQ2 To what extent can the accuracy be explained by spoken content or recording channel rather than accent?

RQ3 Can a training objective that knows the map make errors geographically closer without a large loss in accuracy?

RQ4 Do fine-tuning or temporal architectures add anything over pooled frozen features at this data size?

Our contributions are:

- A frozen-encoder pipeline with a learned layer-weighted head that reaches 64.6% accuracy and 0.637 macro-F1 on the ViMD test set, about 0.23 macro-F1 above the published fine-tuned baselines, which were obtained under a different protocol (Section VI-D).
- A set of confound controls (text-only, spectrum-only and place-name-free evaluations, speaker sex) and channel-perturbation tests. Explicit place names and transcript cues explain little of the accuracy, but the model is sensitive to the long-term spectrum, noise and band-limiting, so accent and recording source cannot be fully separated on ViMD.
- Geo-aware Label Smoothing (Geo-LS), which applies similarity-based label smoothing [2], [3] with a geographic kernel and is closely related to SORD [4] and haversine smoothing [5]. We do not claim the loss form as new; the contribution is its use for dialect geography and a controlled comparison with label smoothing and SORD.

Relative to standard label smoothing, it reduced the mean geographic error by 8 to 26 km in all four settings tested, at a cost of about one point of single-run accuracy on the XLS-R model and no measurable cost on the fused model.
- Negative results with diagnostics for full fine-tuning, LP-FT and frame-level CNN and Transformer heads in the configurations we tried, plus a layer-wise analysis showing that dialect information peaks in the middle layers of both XLS-R and PhoWhisper.

II. Related Work

Vietnamese dialect identification. Earlier Vietnamese work mostly separated the three regional dialects. ViMD [1] introduced province-level labels for 63 provinces and benchmarked fine-tuned wav2vec 2.0, XLS-R, Whisper and PhoWhisper models. PhoWhisper-base was best with 0.411 macro-F1, and XLS-R-300m reached 0.373. The authors also found that geography influences which dialects are confused, which motivates our distance-aware objective.

Pretrained speech representations. wav2vec 2.0 [6] and its multilingual version XLS-R [7] learn speech representations from unlabelled audio. Whisper [8] is trained with weak supervision for transcription; PhoWhisper [9] fine-tunes it on 844 hours of Vietnamese speech, including in-house data from about 26,000 speakers across 63 provinces. The SUPERB benchmark [10] evaluates such models frozen, with a learned weighted sum of layers and a light downstream head; we follow this protocol. Layer-wise studies [11] show that different layers encode acoustic, phonetic and lexical information, so the best layer depends on the task.

Pooling and temporal architectures. For speaker and language recognition, attentive statistics pooling [12] and ECAPA-TDNN [13] are standard ways to summarise frame-level features. We adapt both as heads over frozen frames.

Fine-tuning versus probing. Kumar et al. [14] showed that fine-tuning a pretrained model with a randomly initialised head can distort its features, and proposed LP-FT: train the head first, then fine-tune. We test both strategies.

Distance-aware objectives. Label smoothing [15], [16] replaces one-hot targets with a mixture of the true label and a uniform distribution. Soft ordinal labels (SORD) [4] build targets from a distance between classes, hierarchy-aware losses [17] aim to make "better mistakes", and image geolocation uses "haversine smoothing", which decays the target of each geographic cell with its great-circle distance to the true location [5]. Non-uniform smoothing distributions have also been proposed directly: class-similarity based label smoothing assigns smoothed mass according to class similarity for calibration [3], and similarity-based label smoothing for dialogue generation keeps $1-s$ on the true token and distributes s over other tokens in proportion to embedding similarity [2]. Geo-LS has exactly this structure, with geographic distance in place of embedding similarity: it keeps (up to ε/C) label smoothing's fixed mass on the true class and distributes only the smoothed mass by distance. What distinguishes it from SORD and haversine smoothing is only that the true-class mass does not depend on the kernel width. We therefore regard Geo-LS as an application of existing ideas, and our contribution as the evidence of when it helps.

Table I
ViMD Statistics

Property	Value
Provinces (classes)	63 (pre-2025 boundaries)
Regions	North 25, Central 19, South 19
Utterances	18,949
train / valid / test	15,023 / 1,900 / 2,026
Audio	102.56 h, mean 19.5 s / utt.
Speakers	12,953 (88–309 per province)
Utterances per province	263–363 (median 303)
Chance accuracy	1.6%

III. Dataset

We use ViMD [1] with its official splits. Table I summarises the statistics we measured on the released data.

What the label means. According to the dataset paper [1], the audio consists of interviews with local residents published by the 63 provincial television and broadcasting stations, mostly from news programmes; segments contain a single speaker, and news presenters were filtered out. Each recording is labelled with "the provincial dialect of the sample", which in practice is the province of the station. The paper does not describe a procedure that verified each speaker's origin, and it notes that some residents retain the dialect of a region where they previously lived. The label is therefore a noisy proxy for the speaker's dialect and is, by construction, tied to a broadcaster. ViMD also reports about three times as many male as female speakers. The data are released under CC BY-NC-ND 4.0.

Preprocessing. All audio was resampled to 16 kHz mono and stored as FLAC. The splits are intended to be speaker-disjoint [1]. Checking speaker identifiers (province plus speaker ID), we found no overlap between training and validation or between training and test, but two speakers in both the validation and test sets; we removed their validation utterances, leaving 1,898 validation utterances and fully disjoint splits at the level of the released speaker identifiers. We cannot detect the same person appearing under different identifiers. The test set is untouched, so our numbers use the same test utterances as the original paper. Most speakers contribute only one or two utterances (about 1.5 on average).

Geography. Each province is represented by the coordinates of its capital city, and distances between provinces are great-circle (haversine) distances. The median distance from a province to its nearest neighbour is 44 km and the maximum is 113 km. Coordinates are approximate to about 5 km, which is small compared with these distances.

Transcripts. Each utterance comes with a transcript. We use transcripts only for text-only control baselines and to detect spoken place names, never as input to the main model. The most province-specific words in the transcripts are place names

and local topics, consistent with the news-programme origin of the data.

IV. Methodology

Our system keeps pretrained speech encoders frozen and trains only a small head: (i) encode the waveform with a frozen encoder and keep the hidden states of every layer; (ii) mean-pool each layer over time, using only non-padded frames; (iii) standardise each layer with training-set statistics and combine the layers with learned softmax weights; (iv) classify the 63 provinces with a two-layer MLP (512 hidden units, GELU, dropout 0.3).

A. Frozen Backbones

XLS-R. We use facebook/wav2vec2-xls-r-300m [7] (24 Transformer layers, 1,024 dimensions). A validation-set probe showed that dialect information peaks around layer 5, so we keep the embedding output and layers 1 to 12 (13 states) and drop the rest. Long utterances are cut into 12-second windows; the pooled features of each window are averaged, weighted by frame count, over up to 60 seconds of audio.

PhoWhisper. We use the encoder of vinai/PhoWhisper-large [9] (32 layers, 1,280 dimensions), a Whisper model fine-tuned for Vietnamese ASR. Whisper always receives 30 seconds of padded log-mel input, so we pool only over frames that correspond to real audio. We keep all 33 states.

Fusion. For two backbones, each gets its own layer weights and standardisation; the two pooled vectors are concatenated (2,304 dimensions) before the MLP.

B. Layer-Weighted Pooling

For layer l with pooled vector $\mathbf{h}_l$, training mean $\boldsymbol{\mu}_l$ and standard deviation $\boldsymbol{\sigma}_l$, the utterance representation is

$$\mathbf{z} = \sum_{l=0}^{L} \frac{\exp(w_l)}{\sum_{k=0}^{L} \exp(w_k)} \cdot \frac{\mathbf{h}_l - \boldsymbol{\mu}_l}{\boldsymbol{\sigma}_l}. \tag{1}$$

The weights w are learned jointly with the MLP, as in SUPERB [10]. Because standardisation and mean-pooling are both linear, features can be pooled once and cached, so each head trains in seconds.

C. Geo-Aware Label Smoothing (Geo-LS)

Standard label smoothing, as implemented here, mixes the one-hot target with a uniform distribution over all $C = 63$ provinces, so the true province receives $1-\varepsilon+\varepsilon/C$ (0.705 for $\varepsilon = 0.3$). Geo-LS gives the true province exactly $1-\varepsilon$ (0.700) and spreads ε only over the other provinces, in proportion to their geographic closeness; the difference in true-class mass is negligible. For true province i and distance $d(i, j)$ in kilometres,

$$q_i(j) = \begin{cases} 1-\varepsilon, & j = i, \\ \varepsilon \dfrac{\exp(-d(i,j)/\tau)}{\sum_{k \neq i} \exp(-d(i,k)/\tau)}, & j \neq i. \end{cases} \tag{2}$$

The off-diagonal terms sum to ε, so q_i is a valid distribution for any $\tau > 0$. The model is trained with cross-entropy against q_i. We select ε and τ on the validation set from $\varepsilon \in \{0.1, 0.2, 0.3\}$ and $\tau \in \{25, 50, 100, 200\}$ km; the chosen setting, $\varepsilon = 0.3$ and $\tau = 200$ km, lies at the upper edge of both grids, so larger values were not explored.

SORD [4] and haversine smoothing [5] derive the whole target, including the true class, from the distance kernel. Neighbouring provinces are only about 44 km apart, so a kernel wide enough to spread mass to neighbours also takes mass from the true class; with $\tau = 50$ km, less than half of the target mass remains on it. Geo-LS fixes the true-class mass independently of τ. The kernel uses straight-line distances between capitals, so it ignores natural dialect boundaries such as mountain passes, and provinces in dense regions spread their smoothing mass over more close neighbours than provinces at the edge of the country.

D. Frame-Level Architectures

To test whether temporal patterns such as tone contours add information beyond mean-pooled features, we extract frame-level features from XLS-R (layers combined with fixed weights taken from a linear probe, 1,024 dimensions, 25 frames per second, first 12 seconds) and train five heads on them: (a) mean pooling + MLP (control); (b) mean and standard deviation pooling + MLP; (c) attentive statistics pooling (ASP) [12]; (d) an ECAPA-style head [13] with three dilated Conv1D blocks with squeeze-excitation, followed by ASP (2.5 M parameters); (e) a two-layer Transformer encoder (4 heads, model dimension 256) followed by ASP (1.6 M parameters).

E. Fine-Tuning

We also fine-tune XLS-R end to end with label smoothing, keeping the convolutional feature encoder frozen and using SpecAugment-style time masking [18] ($p = 0.05$). Variant 1 fine-tunes all 24 layers with a randomly initialised head for 6 epochs on 6-second random crops (encoder learning rate 3×10^{-5}, head 10^{-3}). Variant 2 applies LP-FT [14] to the 12-layer encoder: the head is first trained on frozen features, then the whole model is fine-tuned for 12 epochs with a cosine schedule (encoder 2×10^{-5}, head 2×10^{-4}). Each variant was run once.

V. Experimental Setup

Training. Heads are trained with AdamW (learning rate 10^{-3}, weight decay 10^{-3}, batch 256) for up to 80 epochs, with early stopping on validation accuracy (patience 10). Every head configuration is run with seeds 0, 1 and 2; the objective comparison in Table IV uses seeds 0 to 4. Hyperparameters and model choices are selected on the validation set; test results are reported for all configurations. The experiments were, however, carried out in stages and the test set was evaluated after each stage, so later design decisions (for example, fusing the two backbones) may have been influenced

by earlier test results. The reported intervals do not account for this adaptive process.

Baselines and controls. We train logistic-regression or linear-SVM models on (a) MFCC statistics (40 coefficients with deltas, mean and standard deviation, 240 dimensions); (b) the long-term average log-mel spectrum (LTAS, 80 dimensions); (c) TF-IDF word and character n-grams of the transcript; and (d) fine-tuned PhoBERT [19] on word-segmented transcripts. LTAS averages away most phonetic detail but still mixes recording channel, voice characteristics (including speaker sex) and the long-term distribution of speech sounds, so it is a partial proxy for the channel rather than a pure one. Controls (c) and (d) estimate how much spoken content alone reveals to these particular text models.

Metrics. We report accuracy and macro-F1 over the 63 provinces, top-5 accuracy, region accuracy (the predicted province lies in the true North/Central/South region), mean error distance in km (0 for a correct prediction) and the share of predictions within 100 km of the true province.

Single seeds versus ensembles. Two kinds of numbers appear in this paper and should not be mixed. Values written as mean $\pm$ s.d. are averages over the three seeds trained separately. Paired differences with confidence intervals, and all values in Tables V and VII, are computed for three-seed ensembles, whose predicted probabilities are averaged. Ensembling raises accuracy, so a difference between two ensembles need not equal the difference between the corresponding single-seed means; where both are relevant we report both.

Statistics. Confidence intervals use a paired bootstrap with 1,000 to 2,000 resamples; we call a difference significant when its 95% interval excludes zero. They reflect test-set sampling uncertainty for a fixed trained model or ensemble, not variation across training seeds, which is reported separately. The 2,026 test utterances come from 1,344 speakers; 34% of them contribute more than one utterance (at most 7), so utterances are not independent. For the final models (Section VI-D) we therefore resample speakers rather than utterances (cluster bootstrap). This widens accuracy intervals by a factor of 1.27 to 1.35 but leaves every paired conclusion unchanged, because paired differences are much less affected. Intervals for the earlier experiments (Tables II–VI) resample utterances and are therefore slightly too narrow. ViMD does not identify the source programme of each utterance, so clustering by programme was not possible.

Channel perturbations. To probe the dependence on recording conditions, we re-extract XLS-R and PhoWhisper features for all test utterances after one of five perturbations and apply the trained fusion heads unchanged. (i) *LTAS normalisation*: each utterance's long-term average power spectrum (STFT with 512-point frames and 160-sample hop, louder half of the frames, smoothed over 9 bins) is equalised to the mean spectrum of all validation utterances, removing its spectral "colour"; gains are limited to ± 15 dB. (ii) *LTAS swap*: the same operation, but towards the mean spectrum of a randomly chosen other province, computed from validation utterances of different speakers. If the model read a province's channel signature directly, predictions would move to that donor province. (iii) *LTAS own* (control): the same operation towards the mean spectrum of the utterance's true province. It uses the label and serves only as a diagnostic: it applies the same processing as (i) and (ii) but restores the province's average colour, so comparing it with (i) and (ii) separates the effect of that colour from the effect of the processing itself. (iv) *Telephone*: 300–3400 Hz band-pass, 8 kHz resampling and 8-bit μ-law coding. (v) *Noise*: white Gaussian noise at 20 dB SNR. Re-extracting 48 unperturbed utterances reproduced the cached features exactly (cosine similarity 1.000, identical predictions), so the differences come from the perturbations.

Table II
BASELINES ON THE TEST SET (SINGLE RUN; XLS-R USES ALL 25 STATES, THE FIRST 12 S AND CROSS-ENTROPY). ACCURACY WITH 95% BOOTSTRAP CI.

Model	Input	Acc. (%)	F1	Reg. (%)
LTAS + LR	Spectrum	19.7 [18.0, 21.5]	.182	56.1
MFCC stats + LR	Audio	24.6 [22.8, 26.5]	.235	65.4
TF-IDF + SVM	Text	21.9 [20.1, 23.6]	.208	58.1
PhoBERT (fine-tuned)	Text	17.8 [16.2, 19.4]	.133	67.1
XLS-R layer 4, linear	Audio	44.9 [42.7, 47.0]	.443	84.5
XLS-R wsum, linear	Audio	46.6 [44.4, 48.8]	.459	87.9
XLS-R wsum, MLP	Audio	**49.2** [47.2, 51.3]	**.486**	**89.4**

Reproducibility and compute. Code, configuration and all result files are available at https://github.com/stevedang2805/vimd-geo-dialect-id, including the province coordinates and the exact feature-extraction settings. All experiments ran on Google Colab Pro, mostly on an NVIDIA L4 (24 GB). Feature extraction took 15 to 75 minutes per backbone; each head trains in under a minute.

VI. RESULTS

A. Baselines and Confound Controls

Frozen XLS-R features already reach 49.2% accuracy, more than twice any control baseline (Table II). The text-only and spectrum-only models are 11 to 16 times above chance, which confirms that ViMD contains province cues besides accent and that they must be controlled for.

Spoken content. Only 8.9% of test transcripts contain any province or capital-city name, and 7.5% contain the speaker's own province. Masking all place names lowers the TF-IDF model from 21.9% to 17.4%; the remainder comes from local topics and dialect words. Table III evaluates the audio models on subsets where these shortcuts are reduced. Accuracy without place names is within 0.7 points of the full test set for every model. On the 151 utterances that do name the own province, the text model jumps to 72.9%, while the audio models rise by only 1 to 4 points, within their wide confidence intervals. Both observations indicate that explicit place names are not a major source of the audio models' accuracy.

The "text wrong" subset should be read narrowly: it contains utterances whose transcript the TF-IDF model could not exploit, not utterances free of lexical information. A stronger text model, or the language knowledge inside PhoWhisper, might

Table III
TEST ACCURACY (%) ON CONTROL SUBSETS (3-SEED ENSEMBLES; BASELINE: SINGLE RUN). "TEXT WRONG" AND "LTAS WRONG" ARE UTTERANCES MISCLASSIFIED BY THE TF-IDF AND LTAS BASELINES.

Subset	n	XLS-R base	PhoWhisper	Fusion D (LS)
All test	2,026	49.2	58.4	64.2
No place name	1,846	48.8	57.8	63.6
Own province named	151	50.3	62.3	67.6
Text wrong	1,583	45.9	55.5	61.3
LTAS wrong	1,627	44.0	52.1	58.1
Both wrong	1,290	41.0	49.2	55.4
Both text wrong[a]	1,422	45.8	55.4	61.0
Both spectral wrong[b]	1,411	40.2	47.8	53.7
All four wrong	1,016	37.7	45.7	51.2

[a]TF-IDF and PhoBERT both wrong. [b]LTAS and MFCC both wrong.

still extract cues from them. Requiring both text models to fail (1,422 utterances) changes little: the fusion model keeps 61.0%, 95% of its full-test accuracy.

Concern about PhoWhisper. PhoWhisper is trained for transcription, so it could exploit lexical content more than XLS-R. We find no evidence that it does: the share of accuracy it retains on the place-name-free, text-wrong and both-wrong subsets (98.9%, 94.9% and 84.1%) matches the XLS-R baseline (99.3%, 93.4% and 83.4%). Its ASR training data include in-house speech from 63 provinces whose source is not documented [9]; we cannot verify whether it overlaps with ViMD. Such data carry transcripts, not province labels, so overlap would not leak the labels directly. It could still bias the evaluation: an encoder that has been trained on some test recordings or speakers may represent them unusually well, which could inflate the PhoWhisper and fusion results. The pretraining data of XLS-R also include web and broadcast speech whose overlap with ViMD we cannot check. We therefore treat pretraining exposure as an unresolved source of possible optimism, not as excluded.

Recording channel. The LTAS baseline is 12 times above chance, so long-term spectra differ between provinces. Across provinces, LTAS accuracy and XLS-R accuracy are correlated (Spearman $\rho = 0.47$, $p < 0.001$ over 63 provinces); for example, Ninh Bình reaches 73% with LTAS alone. Channel cues therefore plausibly contribute part of the per-province accuracy. On the 1,290 utterances where both the text and the LTAS model fail, the fusion model keeps 55.4% with label smoothing (95% CI 52.7 to 58.1) and 54.4% with Geo-LS, about 35 times chance. Because LTAS is only a partial channel proxy, this shows that the accuracy is not confined to utterances whose long-term spectrum alone identifies the province, not that the model is free of channel information; the perturbation tests in Section VI-D show that it does use long-term spectral shape. When both spectral baselines fail (LTAS and MFCC, 1,411 utterances), the fusion model keeps 53.7% (84% of its full-test accuracy), and on the 1,016 utterances where all four baselines fail it keeps 51.2% (speaker-level CI 47.4 to 54.9), about 32 times chance. The larger drop on spectral subsets is consistent with some reliance on spectral or channel cues, but utterances that defeat every baseline may also be intrinsically harder, so these drops do not measure reliance directly.

Table IV
TRAINING OBJECTIVES (XLS-R, 13 STATES, FIRST 12 S; MEAN ± S.D. OVER 5 SINGLE-SEED RUNS; PARAMETERS SELECTED ON VALIDATION)

Objective	Acc. (%)	F1	Reg. (%)	km	NLL
Cross-entropy	50.3 ± 0.5	.497	88.3	141	2.22
LS (ε=0.3)	**53.2** ± 1.0	**.525**	86.4	159	2.21
SORD (τ=5)	50.2 ± 0.8	.496	88.4	141	2.20
Geo-LS $(0.3, 200)$	51.9 ± 1.0	.513	**89.7**	**126**	**2.01**

Speaker sex. Women make up 29% of the test utterances, and the male share varies across provinces from 45% to 95% (median 73%; $\chi^2 = 690$, 62 degrees of freedom, $p < 10^{-100}$ on training utterances), so sex is associated with province. It carries little province information, however: predicting the most frequent province for each sex gives 2.2% test accuracy (chance 1.6%). Accuracy does not differ significantly between male and female speakers (fusion with Geo-LS: 62.6% versus 67.0%, difference -4.4 points, speaker-level CI -10.2 to 1.6).

B. Training Objectives

Label smoothing gives the highest accuracy, while Geo-LS gives the closest errors and the lowest negative log-likelihood (Table IV). These experiments use frozen XLS-R with 13 states and the first 12 seconds of audio. Geo-LS exceeds cross-entropy in accuracy for each of the five seeds ($+1.6 \pm 1.2$ points) and places errors closer for each seed (-15.2 ± 5.1 km); between five-seed ensembles the differences are $+3.1$ points (speaker-level 95% CI 1.5 to 4.7) and -15.8 km (CI -24.1 to -7.8). Label smoothing exceeds cross-entropy for every seed ($+2.9 \pm 1.2$ points) but moves errors 17.5 km further away on average.

SORD needs a tiny τ. With $\tau = 5$ km the SORD target is almost one-hot, since no neighbour lies that close ($e^{-44/5} < 10^{-3}$), and SORD behaves like cross-entropy (-0.1 ± 1.2 points per seed; five-seed ensembles $+0.4$, CI -0.4 to 1.3). With $\tau = 50$ km on 25 layers it loses 7.7 points of accuracy against cross-entropy (CI -9.6 to -5.7) without reducing error distance; $\tau = 100$, 200 and 400 km are worse still (mean accuracy 35.0%, 27.8% and 24.3%). In this grid, no single kernel width kept the true class dominant while also spreading useful mass to neighbours; Geo-LS separates the two by construction.

Geo-LS versus label smoothing across settings. The geographic gain of Geo-LS is consistent, while its accuracy cost varies and is not significant on the strongest model (Table V).

A non-significant difference does not establish that two models are equivalent. For the fusion model, the one-sided 95% lower bound of the accuracy difference (speaker-level bootstrap) is -1.3 points and the 90% interval is -1.3 to 0.7 points. The data are therefore compatible with accuracy differences between -1.3 and $+0.7$ points. A pre-specified equivalence margin of 1.5 points would have been met and a

Table V
GEO-LS MINUS LABEL SMOOTHING (PAIRED BOOTSTRAP ON ENSEMBLES, 95% CI; ROWS WITHOUT A MARK: 3-SEED ENSEMBLES, UTTERANCE-LEVEL BOOTSTRAP)

Setting	Δ Acc. (points)	Δ Mean error (km)
XLS-R, 25 states	-1.2 $[-2.5, 0.2]$	-12.6 $[-22.1, -3.2]$
XLS-R, 13 states[a]	-0.8 $[-2.1, 0.5]$	-25.7 $[-36.9, -15.6]$
XLS-R frame, mean head	-0.7 $[-2.0, 0.4]$	-18.7 $[-28.6, -9.5]$
Fusion (final)[b]	-0.3 $[-1.5, 0.9]$	-8.2 $[-15.3, -1.2]$

[a]Five-seed ensembles, speaker-level bootstrap. [b]Three-seed ensembles, speaker-level bootstrap.

Table VI
FINE-TUNING AND FRAME-LEVEL HEADS ON XLS-R WITH LABEL SMOOTHING. FROZEN ROWS: FRAME-LEVEL PIPELINE, MEAN ± S.D. OVER 3 SEEDS. FINE-TUNING ROWS: SINGLE RUNS.

Model	Params	Acc. (%)	km	Reg. (%)
Frozen, mean + MLP (ref.)	0.56 M	**52.7** ± 1.1	156	86.4
Frozen, ECAPA-style + ASP	2.5 M	48.0 ± 0.8	141	88.5
Frozen, Transformer + ASP	1.6 M	47.3 ± 0.5	**133**	**89.6**
Full FT, 24 layers, 6 ep.	∼300 M	40.7	244	79.0
LP-FT, 12 layers, 12 ep.	∼150 M	no gain over LP (see text)		

margin of 1 point would not; because no margin was fixed in advance, we do not claim non-inferiority or equivalence.

Seed-to-seed variation. The comparisons above use ensembles. For the 13-state XLS-R model, Geo-LS is below label smoothing for each of the five seeds (-1.3 ± 0.9 points) while its errors are closer for each seed (-32.6 ± 7.8 km): on this model, Geo-LS trades about one point of single-run accuracy for a large geographic gain. Comparing the three fusion heads seed by seed, Geo-LS minus label smoothing gives $+1.8$, 0.0 and -2.0 accuracy points, two of them individually significant in opposite directions, but -14.6, -15.4 and -7.7 km in error distance (mean -12.6 ± 4.2 km). The geographic gain therefore has the same sign for every seed, whereas accuracy differences of up to 2 points between single runs are within seed-to-seed variation. Single-run comparisons elsewhere in this paper that differ by 1 to 2 points should be read with this in mind.

Fewer layers. Keeping only the embedding output and layers 1 to 12 of XLS-R raises the single-seed mean of every objective over using all 25 states: cross-entropy from 49.7% to 50.3%, label smoothing from 51.6% to 53.2%, Geo-LS from 50.4% to 51.9% (three seeds for 25 states, five for 13 states). The gains are small relative to seed variation, but they agree in direction with the layer probe in Section VII-A, which shows that the upper layers carry less dialect information.

C. Fine-Tuning and Frame-Level Architectures

In our configurations, neither fine-tuning nor a more complex head beat a frozen encoder with mean pooling and an MLP (Table VI). The frozen rows of Table VI come from the frame-level pipeline (Section IV-D), which uses fixed layer weights; its mean-pooling reference (52.7%) is therefore not identical to the learned-weight model of Table IV (53.2% with label smoothing).

Full fine-tuning. With a random head, validation accuracy started at 15% after the first epoch and was still rising after epoch 6 (43.5%), so this run was under-trained. Its best test accuracy (40.7%) stayed far below the frozen baseline. A longer schedule might narrow the gap; we did not test it.

LP-FT. Starting from a trained head, the model at epoch 0 reproduces the frozen result (validation 55.7%), which confirms that the pipelines match. One epoch of fine-tuning then drops validation accuracy to 46.0% (37.2% with Geo-LS), and it recovers only to 50.9% after 11 epochs, never reaching the starting point. Training loss kept falling while validation accuracy stayed below the starting point, which is consistent with overfitting; one possible explanation, which we did not test, is that with about 240 utterances per province the encoder adapts to speaker and recording specifics rather than to accent. Parameter-efficient fine-tuning and other learning-rate schedules remain untested.

Temporal heads. Comparing three-seed ensembles, the Conv1D and Transformer heads lose 3.7 and 4.3 points of accuracy against mean pooling (CIs -5.5 to -1.8 and -6.3 to -2.5) but bring errors 17.9 and 26.5 km closer; the corresponding differences between single-seed means in Table VI are 4.7 and 5.4 points and 15 and 23 km. They also raise region accuracy by 2 to 3 points. They capture coarse regional patterns but not the finer distinctions between neighbouring provinces. In an earlier run with 512-dimensional PCA features and 6-second crops, adding standard-deviation statistics hurt further (statistics pooling 42.5%, ASP 38.6%, against 49.0% for mean pooling), possibly because frame-level variance encodes speaker and channel rather than dialect.

Takeaway. With this amount of data, changing the training target brought the geographic benefit of a temporal head at a third of its parameters and without its accuracy cost: in the same frame-level pipeline, mean pooling with Geo-LS reaches 52.2% and 128 km (single-seed means), against 47.3% and 133 km for the Transformer.

D. Backbone Fusion and Final Model

Fusing XLS-R and PhoWhisper is the largest single improvement in this study: $+6.7$ points over XLS-R alone (speaker-level CI 4.9 to 8.6) and $+5.7$ over PhoWhisper alone (CI 3.8 to 7.7). Using the whole utterance instead of the first 12 seconds helps XLS-R only slightly ($+1.1$ points, CI -0.4 to 2.6). Table VII reports three-seed ensembles; the last row averages all nine heads of B, C and D. Single-seed means are about 3 points lower (fusion with LS: $61.1 \pm 1.2\%$; with Geo-LS: $61.0 \pm 0.6\%$).

On the fusion model the geographic gain of Geo-LS is statistically clear but practically modest: 8 km is well below the 44 km median distance between neighbouring provinces, and region accuracy rises only from 93.3% to 94.4%. The gain is larger on weaker models (26 km for XLS-R alone, Table V). We take the fusion architecture with Geo-LS as the main system: it is within 0.7 points of the nine-head ensemble, whose gain over the fusion ensemble is not significant ($+0.4$ points, CI -0.7 to 1.5), and it makes the geographically closest

Table VII
FINAL MODELS ON THE VIMD TEST SET (3-SEED ENSEMBLES; LAST ROW: ALL 9 HEADS OF B, C AND D; 95% SPEAKER-LEVEL CLUSTER BOOTSTRAP CI)

Model	Accuracy (%)	Macro-F1	Top-5 (%)	Region acc. (%)	Mean error (km)	Within 100 km (%)
B: XLS-R, full utterance, LS	57.5 [54.6, 60.3]	0.569 [0.535, 0.588]	81.1	88.9	127	69.2
C: PhoWhisper-large, LS	58.4 [55.7, 61.2]	0.575 [0.544, 0.595]	84.3	93.0	98	71.9
D: Fusion B + C, LS	64.2 [61.2, 66.9]	0.635 [0.603, 0.652]	85.6	93.3	88	75.8
D: Fusion B + C, Geo-LS	63.9 [61.0, 66.6]	0.631 [0.599, 0.651]	86.3	**94.4**	**80**	76.4
Ensemble of B, C and D, LS	**64.6** [61.8, 67.4]	**0.637** [0.605, 0.658]	**86.5**	93.8	82	**76.6**

errors. Against our first frozen XLS-R baseline (49.2%, a single run), it improves accuracy by 14.7 points (CI 12.4 to 17.1) and reduces the mean error by 60 km (CI 50 to 72).

Calibration. Both fusion ensembles are strongly underconfident: their mean top-class probability is 0.31 to 0.32 while accuracy is 0.64, giving expected calibration errors (ECE, 15 bins) of 0.33 with label smoothing and 0.32 with Geo-LS. This is expected with $\varepsilon = 0.3$ smoothing [16], and averaging over seeds lowers confidence further. Geo-LS lowers NLL (1.71 versus 1.83), but its differences in ECE (-0.010, CI -0.021 to 0.003) and Brier score (-0.002, CI -0.007 to 0.003) are not significant. Temperature scaling fitted on the validation set fixes this without changing any prediction: the fitted temperatures (0.47 with label smoothing, 0.46 with Geo-LS) sharpen the distributions and reduce test ECE to 0.047 and 0.031, Brier score to 0.490 for both, and NLL to 1.42 and 1.38.

Robustness to channel perturbations. Table VIII shows that the fusion model depends strongly on the recording conditions. Equalising the long-term spectrum to a neutral average costs 17.0 points of accuracy with label smoothing (speaker-level CI 14.7 to 19.3) and 16.0 with Geo-LS, although region accuracy stays near 90%. Imposing another province's mean spectrum costs about the same, 17.3 points, but moves few predictions to that donor province: 1.7% of test utterances are assigned to it, against 0.5% before the swap (increase 1.2 points, CI 0.7 to 1.8). This is about twice the 0.9% expected if the new errors were spread evenly over the other provinces, so a province's average spectrum exerts a measurable but small pull. The control condition separates two effects. Equalising towards the utterance's own province still costs 7.7 points (CI 5.5 to 9.8), so part of the drop comes from flattening the utterance's individual spectrum and from the processing itself. Accuracy is about 9 points higher with the true province's average spectrum (56.5%) than with a neutral one (47.1%), and the two drops have non-overlapping intervals. We read this difference as an association with province-conditioned spectral information, not as a clean decomposition of the accuracy. This province-average spectrum mixes several sources that our design cannot separate: the broadcaster's channel, the province's mix of male and female voices, and regional differences in voice quality and in the distribution of speech sounds, which are themselves part of dialect. The 9 points therefore measure reliance on province-conditioned spectral information, not on the channel specifically; the comparison also assumes that the processing cost is the same in both conditions. In summary, the model is sensitive to province-conditioned long-term spectral information, whatever its source, but 47% accuracy, about 30 times chance, remains when that information is replaced by a neutral spectrum. Telephone-band audio and white noise at 20 dB SNR are far more damaging (accuracy 16 to 17% and 29 to 31%): the system, trained on clean wideband broadcasts without augmentation, does not transfer to such conditions. Under every perturbation the Geo-LS model keeps its errors closer than the label-smoothing model, for example 155 versus 235 km in noise, with region accuracy 84.6 versus 77.4%.

Table VIII
FUSION MODEL UNDER CHANNEL PERTURBATIONS OF THE TEST AUDIO (3-SEED ENSEMBLES; LS / GEO-LS). DIFFERENCES IN THE TEXT ARE COMPUTED FROM UNROUNDED VALUES.

Condition	Accuracy (%)	Region acc. (%)	Mean error (km)
Clean	64.2 / 63.9	93.3 / 94.4	88 / 80
LTAS own (control)	56.5 / 56.0	89.8 / 92.1	111 / 95
LTAS normalised	47.1 / 47.9	87.2 / 90.0	136 / 114
LTAS swapped	46.8 / 46.7	87.1 / 89.5	139 / 118
Telephone band	16.7 / 15.9	69.3 / 78.0	260 / 228
White noise, 20 dB	28.8 / 31.3	77.4 / 84.6	235 / 155

Comparison with the ViMD paper. Table IX lists macro-F1 together with the fine-tuned baselines published with the dataset. The comparison is not controlled. The paper fine-tunes each model and splits recordings longer than 10 seconds into segments of at most 10 seconds; it does not state how segment predictions are aggregated, so its evaluation unit may differ from our whole utterances. Its best model is PhoWhisper-base (74 M parameters), whereas our systems use frozen XLS-R-300m and the PhoWhisper-large encoder, and their hyperparameters were tuned on our pipeline. The difference therefore combines backbone size, input length, evaluation unit and the frozen-probing protocol, and cannot be attributed to any one of them; in particular, Geo-LS does not improve macro-F1 over label smoothing.

VII. ANALYSIS

A. Where Dialect Lives in the Encoders

Dialect information peaks in the middle of both encoders and fades towards the output (Fig. 1). For XLS-R, a linear probe on single layers is best at layer 5 (51.0% validation accuracy). For PhoWhisper, it is best at layer 16 of 32 (53.4%) and falls to 38.0% at the last layer, which Whisper shapes for transcription. Fig. 1 covers only layers 0 to 12 of XLS-R. An

Table IX
MACRO-F1 ON THE VIMD 63-PROVINCE TEST SET. VIMD ROWS USE A DIFFERENT PROTOCOL (SEE TEXT).

System	Approach	Macro-F1
wav2vec2-large-xlsr-53 [1]	Fine-tuned, $\leq$10 s seg.	0.326
wav2vec2-xls-r-300m [1]	Fine-tuned, $\leq$10 s seg.	0.373
PhoWhisper-base [1]	Fine-tuned, $\leq$10 s seg.	0.411
Ours: XLS-R, 13 states, LS	Frozen; 5-seed mean	0.525
Ours: fusion, Geo-LS	Frozen; 3-seed ens.	0.631
Ours: ensemble of B, C and D	Frozen; 9 heads	**0.637**

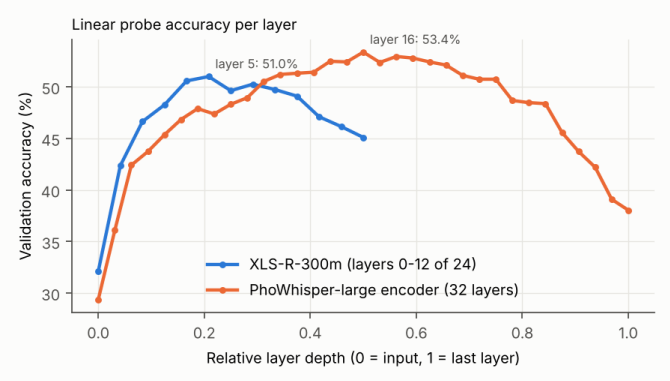


Figure 1. Linear-probe validation accuracy for each layer, plotted against relative depth. Only layers 0 to 12 of XLS-R were extracted.

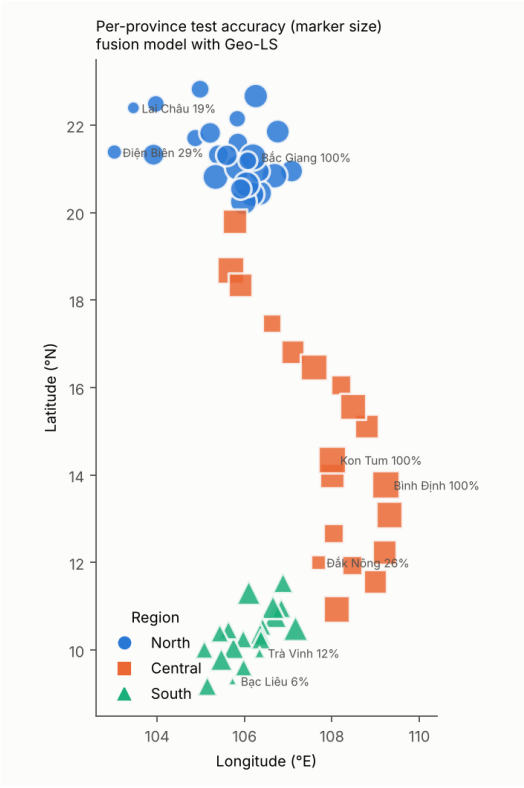


Figure 2. Per-province test accuracy of the fusion model with Geo-LS. Each marker sits at the province capital; its size is proportional to accuracy.

earlier probe over all 25 states, on a balanced subset of 2,520 utterances with its own speaker-disjoint split, peaked at layer 4 (32.1%), fell to 24.0% at layer 12, stayed between 22% and 28% up to layer 22 and dropped to 16% at the last two layers. Because that subset and split differ from the main experiments, it supports the decline only qualitatively; it is consistent with dropping the upper half of XLS-R (Section VI-B). The learned layer weights agree: XLS-R puts most of its weight on layers 3 to 6, and the PhoWhisper-only model on layers 10 to 20. In the fusion model, PhoWhisper also gives noticeable weight to its last two layers, possibly because they carry information that XLS-R lacks.

B. Errors Follow Geography

When the model is wrong, it is usually wrong nearby. For the first XLS-R baseline, trained with plain cross-entropy, wrong predictions were on average 267 km from the true province, against 682 km for a random wrong guess. To test whether confusions decrease with distance, we symmetrise the row-normalised confusion rates over the 1,953 unordered province pairs and correlate them with distance. Pairs that share a province are not independent, so we use a Mantel permutation test [20] that permutes the province labels of the distance matrix (9,999 permutations). For the cross-entropy baseline, Spearman $\rho = -0.42$ ($p \leq 10^{-4}$, the smallest attainable; 95% of permuted values lie between -0.05 and 0.04). The fusion models give $\rho = -0.38$ with label smoothing and -0.41 with Geo-LS, and 76% of test utterances are placed within 100 km of the true province. Because Geo-LS trains the model to prefer nearby provinces, evidence about the data comes from the cross-entropy model. Its pattern is consistent with gradual geographic variation in accent, but neighbouring provinces may also share broadcasting styles or recording conditions, which would produce the same pattern.

C. Easy and Hard Provinces

Per-province accuracy of the fusion model with Geo-LS ranges from 6% to 100% (Fig. 2). The hardest provinces are Bạc Liêu (6%), Trà Vinh (13%), Lai Châu (19%), Đắk Nông (26%) and Điện Biên (29%); Bắc Giang, Bình Định and Kon Tum reach 100%. Each province has only about 32 test utterances, so these values are imprecise. A plausible reading is that the hard provinces sound very close to their neighbours in the Mekong Delta, or have mixed migrant populations in the highlands; we have not verified this. Some of the easiest provinces, such as Ninh Bình, also have a distinctive recording channel, so their scores may be partly inflated.

VIII. LIMITATIONS

Label validity and recording source. Labels denote the province of the broadcasting station, not a verified speaker origin, and each province's audio comes from one broadcaster. Our controls show that explicit place names and transcript cues explain little of the accuracy, and the perturbation tests show that long-term spectral shape matters; neither separates accent from broadcaster-specific recording conditions. Results should

be read as accuracy on ViMD-like broadcast recordings, not as speaker-origin identification from arbitrary audio.

Speaker sex. The sex ratio varies strongly across provinces, but sex alone predicts almost nothing and accuracy does not differ significantly between male and female speakers. The female subset (596 utterances) is small, so differences of a few points cannot be excluded.

Robustness and out-of-domain data. Accuracy falls sharply when the long-term spectrum is equalised (to 47%; restoring the true province's average spectrum recovers about 9 points), with telephone-band audio (16 to 17%) and with 20 dB white noise (29 to 31%). The system is therefore only suitable for clean broadcast-quality audio. We have not evaluated on independently collected recordings from speakers of known origin.

Small data per class. With about 240 training utterances per province, LP-FT and the larger frame-level heads showed signs of overfitting, while the full fine-tuning run was still improving when it stopped after six epochs and was therefore under-trained rather than overfitted. The negative fine-tuning results hold for the two schedules we tried, each run once, not for fine-tuning in general; parameter-efficient methods such as LoRA or adapters remain untested.

Statistics. Speaker-level intervals are available only for the final models; earlier tables use utterance-level intervals, which are slightly too narrow, and no interval accounts for clustering by programme. Intervals reflect test-set uncertainty; seed variability, which is about 1 to 2 accuracy points between single runs, is reported separately. Temperature scaling corrects the under-confidence of the final models on ViMD; whether the fitted temperatures transfer to other audio is untested.

Approximate geography. Each province is reduced to the coordinates of its capital, and labels follow the 63-province boundaries in force before the 2025 merger. The Geo-LS kernel ignores natural dialect boundaries, and its chosen parameters lie at the edge of the search grid.

Test-set reuse. The test set was evaluated after each experimental stage, and the final configuration was chosen with knowledge of those results. Hyperparameters were selected on the validation set and the main comparisons (fusion over single backbones, Geo-LS over label smoothing in error distance) also hold on validation, but the reported test accuracies, especially of the final system, may still be optimistically biased. A confirmatory estimate would require a fresh held-out set, such as newly collected recordings.

Validation–test gap. Validation accuracy is about 6 points above test accuracy for every model, so comparisons between models are consistent, but absolute validation numbers are optimistic. We draw conclusions from test results only.

Comparison with prior work. The ViMD baselines use a different protocol (fine-tuning, smaller backbones, segments of at most 10 seconds), so the gain over them combines several factors.

IX. Conclusion

Frozen pretrained speech encoders with a learned layer-weighted head identify the broadcasting province of Vietnamese speech in ViMD with 63.9% accuracy for a three-seed ensemble of a fused XLS-R and PhoWhisper head (61.0% for individual seeds), and 64.6% for a nine-head ensemble. This is well above the 0.411 macro-F1 published with the dataset, although the protocols differ. Controls with transcripts, spectra, place-name-free subsets and speaker sex show that the accuracy is largely retained when place names are absent or the baselines fail, and does not differ significantly by speaker sex; these controls reduce, but do not rule out, reliance on content and recording source. The accuracy is, however, sensitive to recording conditions: after equalisation, restoring the province-average long-term spectrum recovers about 9 points, and noise and telephone-band audio reduce accuracy sharply, so the system should be regarded as a classifier of broadcast recordings. Geo-aware Label Smoothing reduced the geographic error of every model we tried by 8 to 26 km against standard label smoothing, also kept errors closer under every channel perturbation, and cost about one point of single-run accuracy on the XLS-R model; on the strongest model the accompanying accuracy change was -0.3 points, with a 90% interval inside ± 1.5 points, and the geographic gain had the same sign for every training seed. In this low-resource regime and in our configurations, the frozen-feature approach outperformed full fine-tuning, LP-FT and temporal CNN and Transformer heads.

Four directions follow. First, testing on independently recorded speech from speakers of known origin would measure how far the system generalises beyond ViMD's broadcast conditions. Second, training with noise, band-limiting and random spectral equalisation as data augmentation is the most direct way to address the sensitivity found in our robustness tests. Third, adversarial training against speaker or channel information could reduce the remaining confound. Fourth, parameter-efficient fine-tuning, and joint dialect identification and speech recognition using ViMD's transcripts, could show whether adaptation helps once overfitting is controlled.

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